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(54) **SYSTEMS AND METHODS FOR ADJUSTING ELEVATOR LOAD SETTINGS**

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B66B 1/24 (2006.01)

B66B 1/28 (2006.01)

(52) **U.S. Cl.**

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(58) **Field of Classification Search**

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See application file for complete search history.

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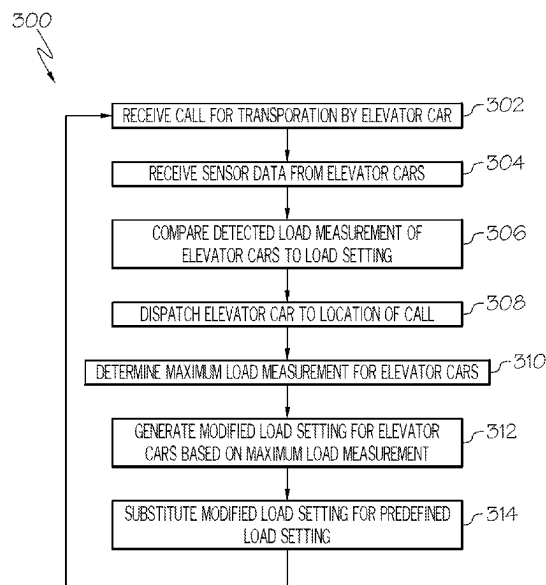
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(57)

ABSTRACT

A method of adjusting a load setting of an elevator car that includes receiving one or more load measurements associated with the elevator car and determining a maximum load of the elevator car from the one or more load measurements. The method further includes generating a modified load setting for the elevator car based on the maximum load and replacing the load setting of the elevator car with the modified load setting.

19 Claims, 5 Drawing Sheets



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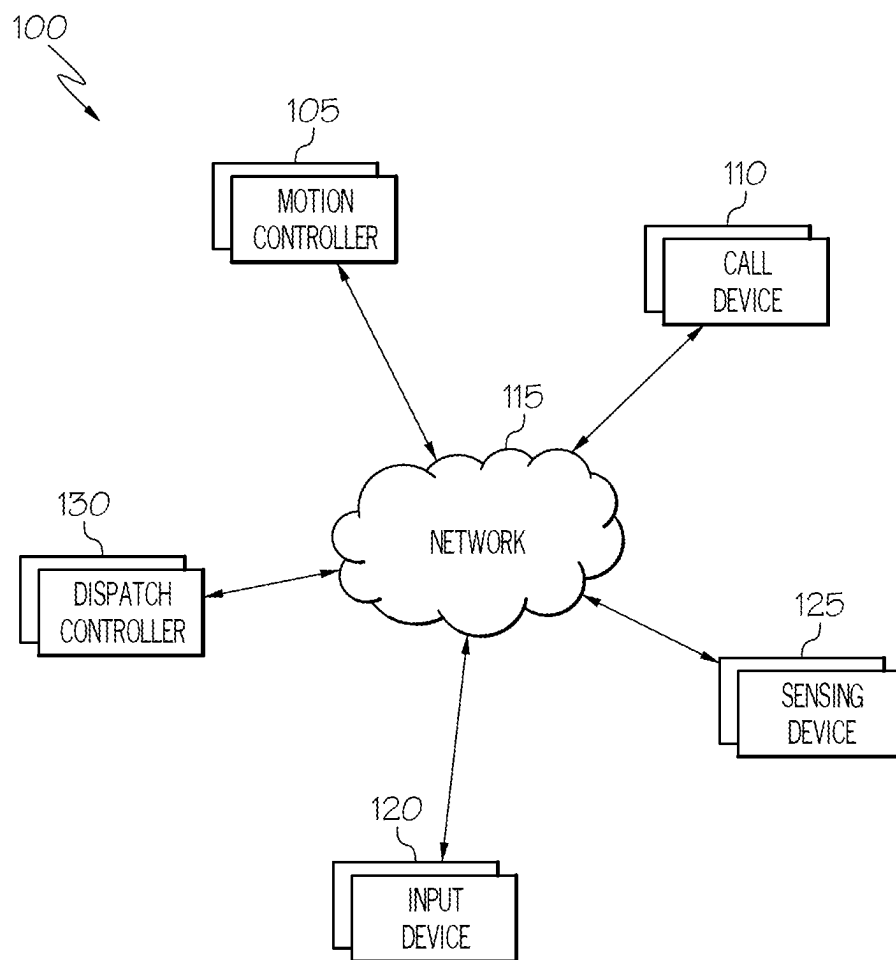


FIG. 1

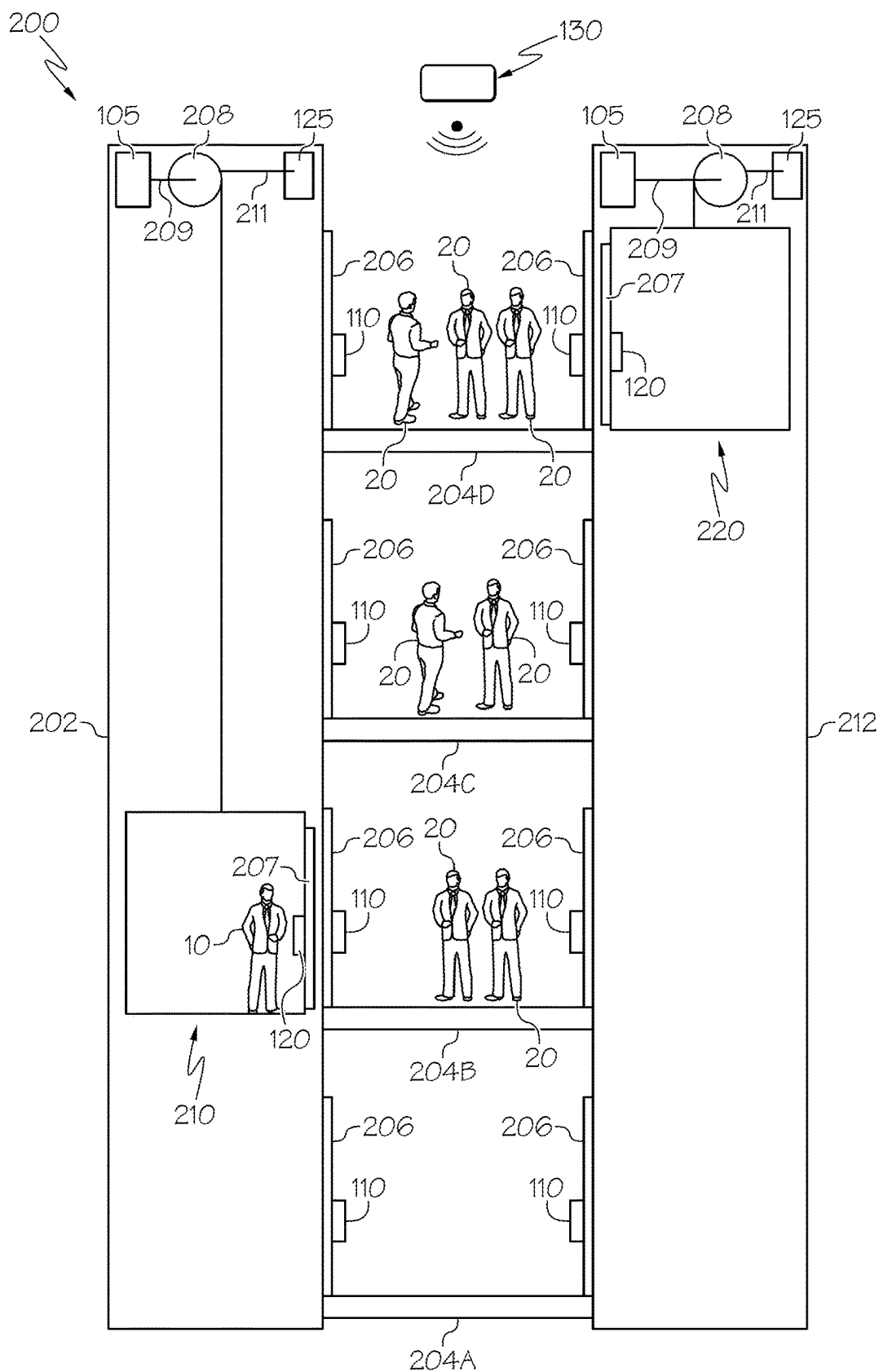


FIG. 2

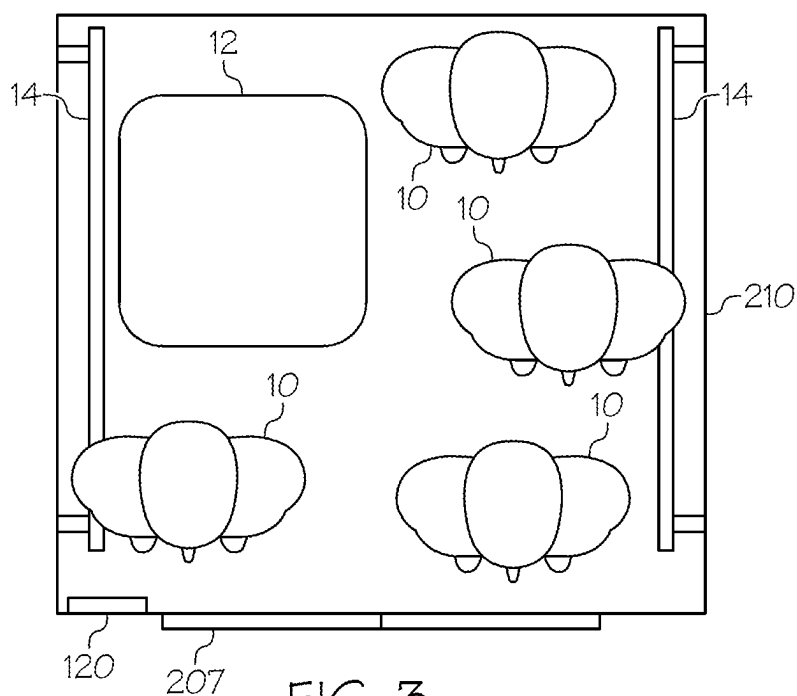


FIG. 3

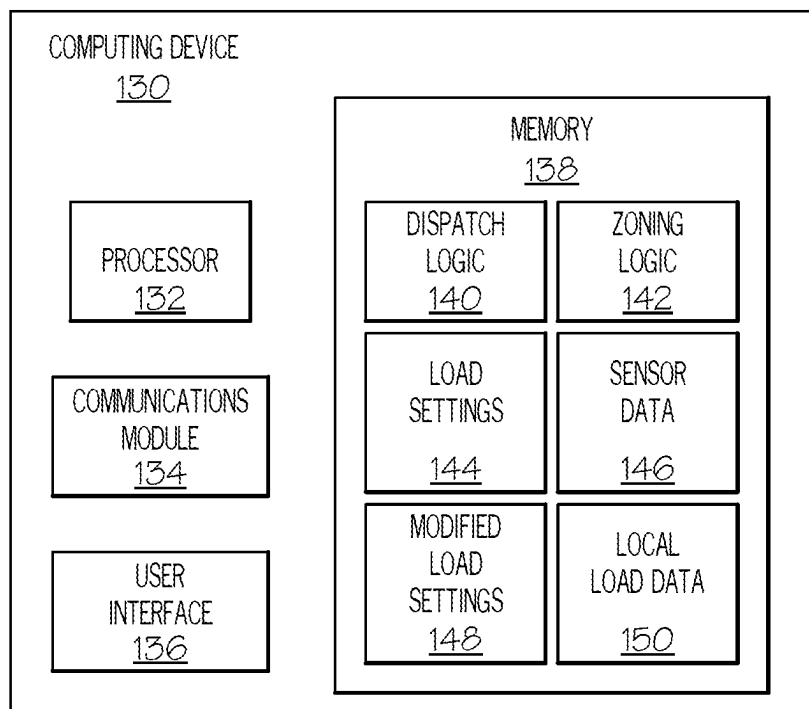


FIG. 4

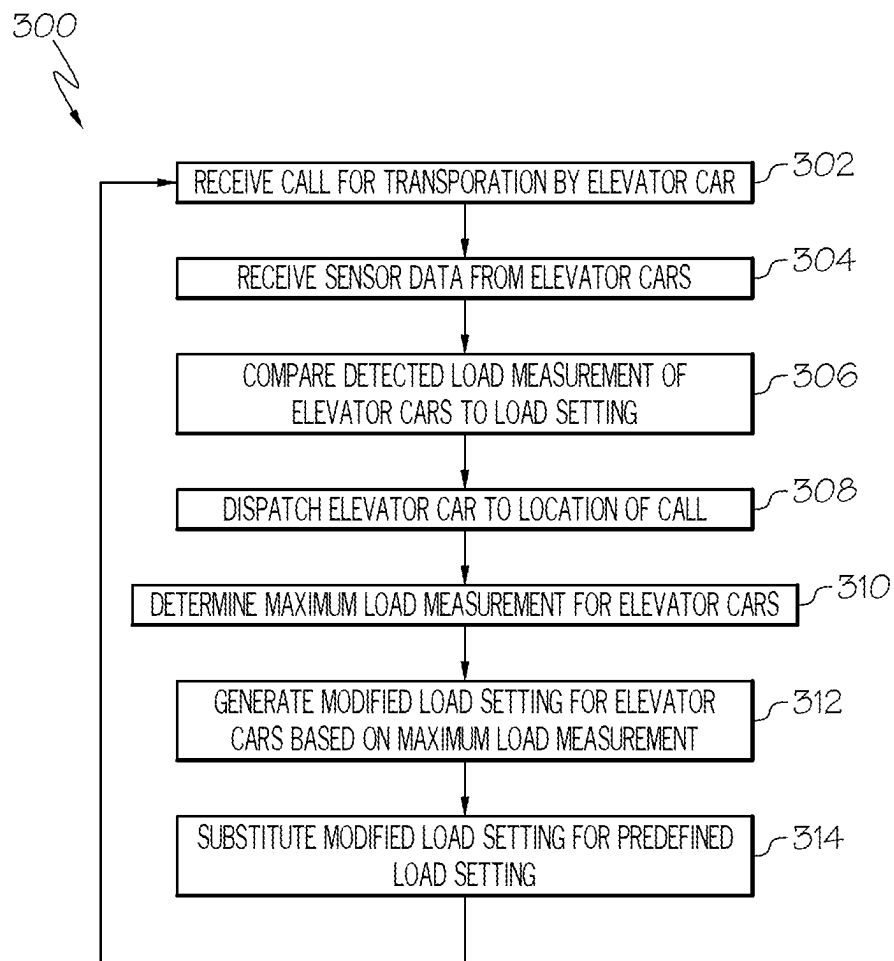


FIG. 5

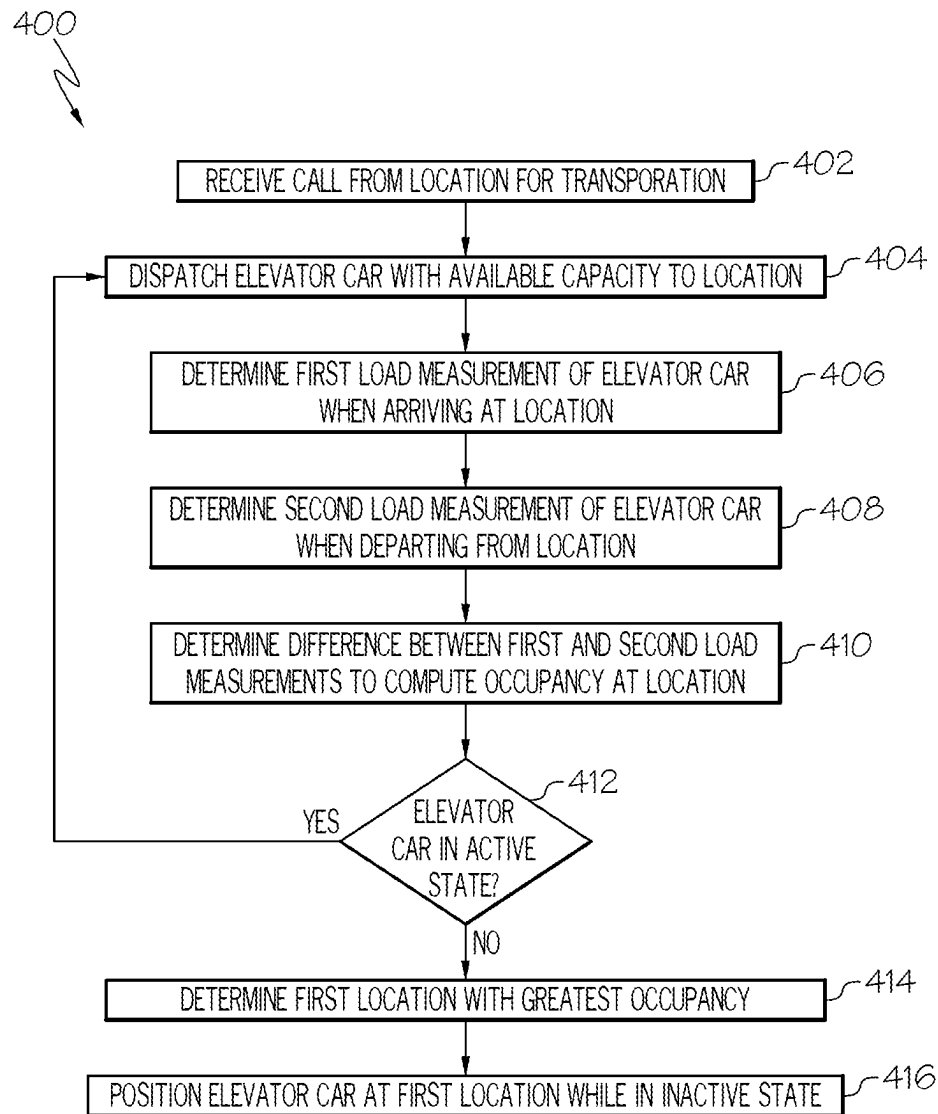


FIG. 6

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SYSTEMS AND METHODS FOR ADJUSTING ELEVATOR LOAD SETTINGS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of priority from U.S. Provisional Application No. 63/070,655, filed on Aug. 26, 2020, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

Aspects of the present disclosure relate generally to systems and methods for controlling elevator traffic flow, and specifically to examples of elevator control systems that dynamically adjust a load capacity setting of elevators based on a maximum detected load.

DESCRIPTION OF RELATED TECHNOLOGY

Elevator systems may generally preset a load capacity setting for elevator cars, which defines a maximum load that each elevator car may receive. The load capacity setting may be preset by a manufacturer of the elevator system or a user of the elevator system. In such systems, an elevator car having a current load that exceeds the preset load capacity may be ignored from consideration for calls from prospective passengers. However, prospective passengers may commonly forgo entering an elevator car that has a current load below the present load capacity for various reasons. For example, prospective passengers may prefer to enter elevator cars having a certain number of occupants that is less than the load capacity of the elevator car. As a result, an elevator car having a current load below the preset load capacity may be dispatched to a location of a prospective passenger but not occupied by the prospective passenger, thereby resulting in decreased traffic flow and greater wait times for prospective passengers who request another elevator car to be dispatched. Providing a system capable of dynamically adjusting a load capacity setting may result in dispatching elevator cars with a greater likelihood of receiving passengers, thereby increasing traffic flow and decreasing wait times for prospective passengers.

BRIEF DESCRIPTION OF DRAWINGS

The accompanying drawings, which are incorporated in and constitute a part of this disclosure, illustrate various exemplary embodiments and together with the description, serve to explain the principles of the disclosure.

Aspects of the disclosure may be implemented in connection with embodiments illustrated in the attached drawings. These drawings show different aspects of the present disclosure and, where appropriate, reference numerals illustrating like structures, components, materials and/or elements in different figures are labeled similarly. It is understood that various combinations of the structures, components, and/or elements, other than those specifically shown, are contemplated and are within the scope of the present disclosure. There are many aspects and embodiments described herein. Those of ordinary skill in the art will readily recognize that the features of a particular aspect or embodiment may be used in conjunction with the features of any or all of the other aspects or embodiments described in this disclosure.

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FIG. 1 depicts a dispatch system including one or more devices in communication over a network.

FIG. 2 is a schematic view of a working environment including multiple elevator cars interacting with the dispatch system shown in FIG. 1.

FIG. 3 is a top view of an interior of an elevator car from the working environment shown in FIG. 2.

FIG. 4 is a schematic view of hardware components of a computing device from the dispatch system shown in FIG. 1.

FIG. 5 is a flow diagram of an exemplary method of adjusting a load setting of elevator cars with the dispatch system shown in FIG. 1.

FIG. 6 is a flow diagram of an exemplary method of positioning inactive elevator cars with the dispatch system shown in FIG. 1.

SUMMARY

According to an example, a method of adjusting a load setting of an elevator car includes receiving one or more load measurements associated with the elevator car and determining a maximum load of the elevator car from the one or more load measurements. The method further includes generating a modified load setting for the elevator car based on the maximum load and replacing the load setting of the elevator car with the modified load setting for a predefined period.

According to another example, a method of operating a plurality of elevator cars includes measuring a load of each of the plurality of elevator cars during a predefined period and determining a maximum load of each of the plurality of elevator cars from the load measurements. The method further includes generating a modified load setting for each of the plurality of elevator cars based on the respective maximum load of each of the plurality of elevator cars, and applying the modified load setting of each of the plurality of elevator cars in place of a load setting during the predefined period. The modified load setting defines an adjusted capacity of each of the plurality of elevator cars relative to the load setting.

According to a further example, a method for positioning an elevator car includes determining an occupancy of each of a plurality of locations by determining a first load measurement of the elevator car upon arriving at each of the plurality of locations, determining a second load measurement of the elevator car upon departing from each of the plurality of locations, and determining a difference between the first load measurement and the second load measurement. The method further includes moving the elevator car to a first location with a total occupancy that is greater than the occupancy at each respective location of the plurality of locations when the elevator car is in an inactive state.

DETAILED DESCRIPTION

The dispatch system of the present disclosure may be in the form of varying embodiments, some of which are depicted by the figures and further described below.

Both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the features, as claimed. As used herein, the terms “comprises,” “comprising,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent

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to such a process, method, article, or apparatus. Additionally, the term “exemplary” is used herein in the sense of “example,” rather than “ideal.” It should be noted that all numeric values disclosed or claimed herein (including all disclosed values, limits, and ranges) may have a variation of $\pm 10\%$ (unless a different variation is specified) from the disclosed numeric value. Moreover, in the claims, values, limits, and/or ranges mean the value, limit, and/or range $\pm 10\%$.

FIG. 1 shows an exemplary dispatch system **100** that may include motion controller **105**, call device **110**, input device **120**, sensing device **125**, and dispatch controller **130**. The one or more devices of dispatch system **100** may communicate with one another across a network **115** and in any arrangement. For example, the devices of dispatch system **100** may be communicatively coupled to one another via a wired connection, a wireless connection, or the like. In some embodiments, network **115** may be a wide area network (“WAN”), a local area network (“LAN”), personal area network (“PAN”), etc. Network **115** may further include the Internet such that information and/or data provided between the devices of dispatch system **100** may occur online (e.g., from a location remote from other devices or networks coupled to the Internet). In other embodiments, network **115** may utilize Bluetooth® technology and/or radio waves frequencies.

Motion controller **105** may be operably coupled to a transportation unit and configured to detect and transmit motion data of the transportation unit to one or more devices of dispatch system **100**, such as, for example, dispatch controller **130**. For example, motion controller **105** may measure and record one or more parameters (e.g., motion data) of the transportation unit, including, but not limited to, a current location, a travel direction, a travel speed, a door location, a status (e.g., active, inactive, moving, parked, idle, etc.), and more. Motion controller **105** may include a computing device having one or more hardware components (e.g., a processor, a memory, a sensor, a communications module, etc.) for generating, storing, and transmitting the motion data. As described in further detail herein, motion controller **105** may be operably coupled to an elevator car located within a building, and dispatch system **100** may include at least one motion controller **105** for each elevator car.

Still referring to FIG. 1, call device **110** may be positioned outside the transportation unit and configured to receive a user input from one or more prospective occupants for accessing the transportation unit. For example, the user input may be indicative of a call requesting transportation from the transportation unit. Call device **110** may be configured to transmit the call request to one or more devices of dispatch system **100**, such as, for example, dispatch controller **130**. Call device **110** may include a keypad, a touchscreen display, a microphone, a button, a switch, etc. Call device **110** may be further configured to receive a user input indicative of a current location of the call request (e.g., a first location) and/or a destination location (e.g., a second location) from a plurality of locations.

As described in further detail herein, call device **110** may be located within a building, and dispatch system **100** may include at least one call device **110** for each floor of the building. Call device **110** may be configured to transmit a message from one or more devices of dispatch system **100** (e.g., dispatch controller **130**) identifying an elevator car designated to arrive at the floor of the building to answer the call request. The message may be communicated by call

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device **100** via various suitable formats, including, for example, in a written form, an audible form, a graphic form, and more.

Input device **120** may be positioned inside the transportation unit and configured to receive a user input from one or more occupants of the transportation unit. For example, the user input may be indicative of a command requesting redirection of the transportation unit. Input device **120** may be configured to transmit the command to one or more devices of dispatch system **100**, such as, for example, dispatch controller **130**. Input device **120** may include a keypad, a touchscreen display, a microphone, a button, a switch, etc. As described in detail herein, input device **120** may be located within an elevator car, and dispatch system **100** may include at least one input device **120** for each elevator car in a building. In other embodiments, input device **120** may be omitted entirely from dispatch system **100**.

Still referring to FIG. 1, sensing device **125** may be positioned inside or outside the transportation unit, and configured to detect and transmit sensor data associated with the transportation unit to one or more devices of dispatch system **100**, such as, for example, dispatch controller **130**. For example, sensing device **125** may measure and record a current load of the transportation unit, including, but not limited to, a weight measurement, a voltage direct current, and more. Sensing device **125** may include a load weighing device, such as, for example, a crosshead deflection device, a rope tension device, a platform movement device, a load sensor or cell (e.g., force transducer), etc. As described in further detail herein, sensing device **125** may be coupled to an elevator car that is located within a building, and dispatch system **100** may include at least one sensing device **125** for each elevator car of the building.

Dispatch controller **130** may be positioned outside the transportation unit and configured to receive data (e.g., motion data, a call request, a redirection command, sensor data, etc.) from one or more devices of dispatch system **100**. Dispatch controller **130** may be configured to determine at least one transportation unit of a plurality of transportation units to dispatch to a location of a call request received from a prospective occupant seeking transportation. Dispatch controller **130** may be further configured to determine a current load of a transportation unit based on the data received from the one or more devices of dispatch system **100**. Dispatch controller **130** may include a computing device (see FIG. 4) operable to perform one or more processes (see FIG. 5) for dynamically adjusting a load setting of transportation units and rendering said transportation units inoperable to receive a call request when a current load exceeds the load setting. Dispatch controller **130** may be further operable to perform one or more processes (see FIG. 6) for moving transportation units in an inactive state to a location with a total occupancy that is greater than an occupancy at a plurality of other locations. As described in further detail herein, dispatch controller **130** may be operably coupled to a plurality of elevator cars located within a building, and dispatch system **100** may include at least one dispatch controller **130** for each building.

Referring now to FIG. 2, dispatch system **100** may be utilized in a working environment **200**, such as a building (e.g., a facility, a factory, a store, a school, a house, an office, and various other structures). In the example, the transportation unit may include one or more elevator cars within the building. It should be appreciated that working environment **200** is merely illustrative such that dispatch system **100** may

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be utilized in various other suitable environments than those shown and described herein without departing from a scope of this disclosure. For example, the working environment may include a mass transit system such that the transportation unit(s) may include a bus, a train, a subway car, a metro car, a vehicle, etc. In the present example, working environment 200 may include a plurality of floors defining a plurality of locations within the building, such as first floor 204A, second floor 204B, third floor 204C, and fourth floor 204D. It should be appreciated that, in other embodiments, the building of working environment 200 may include additional and/or fewer floors.

Working environment 200 may further include one or more elevator shafts with at least one elevator car positioned within each elevator shaft. In the example, working environment 200 includes a first elevator shaft 202 with a first elevator car 210 and a second elevator shaft 212 with a second elevator car 220. Although not shown, it should be appreciated that working environment 200 may include additional (e.g., a plurality) elevator shafts and/or elevator cars. Each elevator car 210, 220 may be coupled to a pulley system 208 configured to move elevator cars 210, 220 within elevator shafts 202, 212 and relative to floors 204A-204D. It should be understood that pulley system 208 may include various mechanical and/or electrical mechanisms for moving elevator cars 210, 220 within elevator shafts 202, 212, including but not limited to, a motor, a cable, a counterweight, a sheave, etc.

Still referring to FIG. 2, each elevator car 210, 220 may include at least one motion controller 105 operably coupled to pulley system 208, such as, for example, via a wireless connection and/or a wired connection 209. Motion controller 105 may be configured to measure motion data (e.g., a status) from elevator cars 210, 220 by detecting a relative movement of pulley system 208. Each elevator car 210, 220 may further include at least one input device 120 positioned within a cabin of elevator car 210, 220 for receiving a user input from one or more occupants 10 located within the cabin.

Each floor 204A-204D may include one or more call devices 110 and access doors 206 providing accessibility to elevator cars 210, 220 when an elevator door 207 of elevator car 210, 220 is aligned with the respective floor 204A-204D. Call device 110 may be configured to receive a user input from one or more prospective occupants 20 located at one of the plurality of floors 204A-204D. For example, call device 110 may be configured to receive a user input indicative of a call requesting transportation via at least one of elevator cars 210, 220. Call device 110 may be configured to transmit the call request to dispatch controller 130, which may include data indicative of a current location within working environment 200 from which the call request originated from. The call request may further include data indicative of a destination location within working environment 200 to which the prospective passenger is seeking transportation to.

Still referring to FIG. 2, each elevator car 210, 220 may further include at least one sensing device 125. Sensing device 125 may be coupled to elevator car 210, 220 and configured to detect a load (e.g., weight) of elevator car 210, 220. With elevator car 210, 220 including one or more occupants 10 within a cabin of elevator car 210, 220, sensing device 125 may be operable to correlate the detected load measurement to a number of occupants 10 within elevator car 210, 220. In some embodiments, sensing device 125 may be positioned on elevator car 210, 220 (e.g., within the cabin). In other embodiments, sensing device 125 may be positioned external to elevator car 210, 220 and coupled to

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pulley system 208. For example, sensing device 125 may include one or more connections 211 coupled to one or more components of pulley system 208 (e.g., a crosshead, a beam, a hitch, a rope, a platform, etc.).

As seen in FIG. 3, sensing device 125 may be configured to measure a total load of elevator cars 210, 220, including any items present within the cabin, and occupying a capacity, of elevator cars 210, 220 (e.g., occupants 10, ancillary objects 12, etc.). In some embodiments, sensing device 125 may detect a total load of elevator cars 210, 220, including a weight of elevator car 210, 220 and the one or more components of elevator car 210, 220 (e.g., rails 14, input device 120, doors 207, etc.). In other embodiments, sensing device 125 may detect a current load of elevator cars 210, 220 in exclusion of any items within the cabin that may not occupy a capacity of elevator cars 210, 220 (e.g., rails 14, input device 120, doors 207, etc.). Sensing device 125 may detect one or more load measurements of elevator cars 210, 220 and record such measurements as sensor data. As discussed further herein, sensing device 125 may be configured to transmit the sensor data for each elevator car 210, 220 to dispatch controller 130 via network 115 for determining an availability of the elevator car 210, 220 to receive prospective passengers 20 from one or more floors 204A-204D.

Referring now to FIG. 4, dispatch controller 130 may include a computing device incorporating a plurality of hardware components that allow dispatch controller 130 to receive data (e.g., motion data, call requests, commands, sensor data, etc.), process information (e.g., current load measurements, load settings, etc.), and/or execute one or more processes (see FIGS. 5-6). Illustrative hardware components of dispatch controller 130 may include at least one processor 132, at least one communications module 134, a user interface 136, and at least one memory 138. In some embodiments, dispatch controller 130 may include a computer, a mobile user device, a remote station, a server, a cloud storage, and the like. In the illustrated embodiment, dispatch controller 130 is shown and described herein as a separate device from the other devices of dispatch system 100, while in other embodiments, one or more aspects of dispatch controller 130 may be integrated with one or more of the other devices of dispatch system 100. Stated differently, the illustrative hardware components of dispatch controller 130 shown and described herein may be integral with one or more of motion controller 105, call device 110, input device 120, and/or sensing device 125.

Processor 132 may include any computing device capable of executing machine-readable instructions, which may be stored on a non-transitory computer-readable medium, such as, for example, memory 138. By way of example, processor 132 may include a controller, an integrated circuit, a microchip, a computer, and/or any other computer processing unit operable to perform calculations and logic operations required to execute a program. As described in detail herein, processor 132 is configured to perform one or more operations in accordance with the instructions stored on memory 138, such as, for example, dispatch logic 140, zoning logic 142, and the like. Communications module 134 may facilitate communication between dispatch controller 130 and the one or more other devices of dispatch system 100, such as, for example, via network 115. User interface 136 may include one or more input and output devices, including one or more input ports and one or more output ports. User interface 136 may include, for example, a keyboard, a mouse, a touchscreen, etc., as input ports. User interface 136 may further include, for example, a monitor, a display, a

printer, etc. as output ports. User interface 136 may be configured to receive a user input indicative of various commands, including, but not limited to, a command to execute one or more processes (FIGS. 5-6), a command defining a predefined period, a command to apply an automatic adjustment of a load setting, and more.

Still referring to FIG. 4, memory 138 may include various programming algorithms and data that support an operation of dispatch system 100. Memory 138 may include any type of computer readable medium suitable for storing data and algorithms, such as, for example, random access memory (RAM), read only memory (ROM), a flash memory, a hard drive, and/or any device capable of storing machine-readable instructions. Memory 136 may include one or more data sets, including, but not limited to, motion data received from motion controller 105, a load setting 144 for each of the plurality of elevator cars 210, 220, sensor data 146 captured from each sensing device 125, a modified load setting 148 for each of the plurality of elevator cars 210, 220, local load data 150, and the like.

Load settings 144 may include data indicative of a preset maximum load capacity for each of the plurality of elevator cars 210, 220. That is, load settings 144 may define a maximum weight that each elevator car 210, 220 may receive during use. It should be appreciated that the load settings 144 for each of the plurality of elevator cars 210, 220 may be the same, or vary, relative to one another. Load settings 144 may be predefined by, for example, a user of dispatch system 100 (e.g., via user interface 136). In some embodiments, load settings 144 may be modified by the user. Sensor data 146 may include a real-time load measurement of each elevator car 210, 220, indicative of a number of occupants 10 (and/or ancillary objects 12) located within a cabin of elevator cars 210, 220. In some embodiments, the sensor data 146 stored in memory 138 may include a maximum load measurement of a respective elevator car 210, 220 detected by sensing device 125. As described in detail herein, modified load settings 148 may include an updated load setting (e.g., maximum load capacity) for each of the plurality of elevator cars 210, 220 based on data received from the one or more devices of dispatch system 100 (e.g., sensing device 125). Dispatch controller 130 may be configured to dynamically generate the modified load settings 148 based on one or more load measurements received from sensing devices 125 of elevator cars 210, 220.

Still referring to FIG. 4, the modified load settings 148 may further include a predefined period during which the modified load settings 148 may be applied by dispatch controller 130. Dispatch controller 130 may be configured to replace the load settings 144 with the modified load settings 148 during the predefined period. In some embodiments, dispatch controller 130 may be configured to autonomously determine the predefined period, while in other embodiments a user of dispatch system 100 may manually select the predefined period (e.g., via user interface 136).

Local load data 150 may include a load balance measurement at each of the plurality of locations within working environment 200, and may be indicative of a number of occupants 10 located at each of the plurality of floors 204A-204D. Dispatch controller 130 may be configured to compute the local load data 150, which may correspond to a load of items (e.g., occupants 10, ancillary objects 12, etc.) transported to and from each of the plurality of floors 204A-204D by at least one of the plurality of elevator cars 210, 220. Dispatch controller 130 may be further configured to store the local load data 150 in memory 138 and associate the load with a number of occupants located at a particular

location within working environment 200 (e.g., floors 204A-204D). For example, dispatch controller 130 may receive and correlate the motion data received from motion controller 105 with the sensor data 146 received from sensing device 125 to determine the local load data 150.

In some embodiments, dispatch controller 130 may be configured to periodically (e.g., hourly, daily, weekly, monthly, yearly, etc.) update the modified load settings 148 for each of the plurality of elevator cars 210, 220 based on receiving additional load measurements (e.g., sensor data 146) from sensing devices 125. In further embodiments, dispatch controller 130 may be further configured to periodically update the local load data 150 upon determining one or more elevator cars 210, 220 have traveled to and/or from floors 204A-204D to transport at least one occupant 10. That is, dispatch controller 130 may continuously modify the local load data 150 to include a current load balance measurement at each floor 204A-204D based on determining a number of occupants 10 arriving to, or leaving from, each floor 204A-204D (e.g., as detected by sensing device 125).

Still referring to FIG. 4, memory 138 may include a non-transitory computer readable medium that stores machine-readable instructions thereon, such as, dispatch logic 140 and zoning logic 142. In one example, dispatch logic 140 may include executable instructions that allow dispatch system 100 to determine an occupant capacity of each elevator car 210, 220 based on a current load measurement of each elevator car 210, 220 (e.g., sensor data 146). As described in detail herein, dispatch system 100 may be configured to determine whether a current load of each elevator car 210, 220 (indicative of a number of occupants present within the cabin) exceeds a maximum load capacity of the respective elevator car 210, 220 (e.g., load setting 144, modified load setting 148). When the maximum load capacity of at least one elevator car 210, 220 is exceeded, dispatch system 100 may render the particular elevator car inoperable to answer additional call requests from prospective occupants 20 seeking transportation. That is, dispatch system 100 disregards the elevator car from further consideration when determining which of the plurality of elevator cars 210, 220 to dispatch to a new call request(s) until the current load of the elevator car no longer exceeds the maximum load capacity.

In another example, zoning logic 142 may include executable instructions that allow dispatch system 100 to determine when one or more of the plurality of elevator cars 210, 220 is in an inactive state, and which location (e.g., a first location) to park elevator cars at while in the inactive state. The executable instructions of zoning logic 142 may further allow dispatch system 100 to determine an amount of load transferred by elevator cars 210, 220 to a plurality of locations (e.g., floors 204A-204D) to identify a first location having a greater load balance than the remaining plurality of locations.

Referring now to FIG. 5, an example method 300 of using dispatch system 100 to dynamically adjust a load setting of an elevator car, and to render the elevator car inoperable for receiving calls when a current load exceeds the load setting, is depicted. It should be understood that the steps shown and described herein, and the sequence in which they are presented, are merely illustrative such that additional and/or fewer steps may be included in various arrangements without departing from a scope of this disclosure.

At step 302, dispatch system 100 may receive a call at a location of a plurality of locations within working environment 200. The call may be initiated in response to a

prospective occupant **20** actuating call device **110** at the location (e.g., a first location). Call device **100** may transmit the call to dispatch controller **130** via network **115**, and the call may include data indicative of the first location (e.g., fourth floor **204D**) from which the call originated from. The call may further include data indicative of a destination (e.g., a second location) within working environment **200** to which the prospective occupant **20** seeks to travel, such as first floor **204A**.

Dispatch controller **130**, in accordance with dispatch logic **140**, may retrieve motion data of each elevator car **210**, **220**, from a corresponding motion controller **105**, to determine movement parameters of elevator cars **210**, **220**. For example, dispatch controller **130** may receive data including a current location, a travel direction, a travel speed, etc., of each elevator car **210**, **220**. Dispatch controller **130** may further retrieve a load measurement (e.g., sensor data **146**) of each elevator car **210**, **220**, from a corresponding sensing device **125**, at step **304**. Dispatch controller **130** may be configured to determine a current load of each of elevator cars **210**, **220** based on the sensor data **146**.

Still referring to FIG. **5**, at step **306**, dispatch controller **130** may compare the current load measurement of each elevator car **210**, **220** to a respective load setting **144** to determine whether the current load exceeds a maximum load capacity (e.g., the load setting **144**) of the elevator car **210**, **220**. Load setting **144** may include various suitable capacities, including, but not limited to, a range of about 1,000 pounds to about 3,000 pounds. In the present example, the load setting **144** of first elevator car **210** may be about 1,500 pounds, and the load setting **144** of second elevator car **220** may be about 1,400 pounds. Dispatch controller **130** may be configured to analyze the motion data and the sensor data **146** of the plurality of elevator cars **210**, **220** to determine which elevator car **210**, **220** to dispatch to the first location, at step **308**.

For example, in response to determining the current load does not exceed the load setting **144**, dispatch controller **130** may be configured to render the elevator car **210**, **220** operable to receive the call. That is, dispatch controller **130** may determine the elevator car **210**, **220** is available for consideration when determining which of the plurality of elevator cars **210**, **220** to dispatch to the call request. In response to determining the current load exceeds the load setting **144**, dispatch controller **130** may be configured to render the elevator car **210**, **220** inoperable to receive the call. In this instance, dispatch controller **130** may determine the elevator car **210**, **220** is unavailable such that the elevator car **210**, **220** is omitted from consideration when determining which of the plurality of elevator cars **210**, **220** to dispatch to the call.

In the present example, first elevator car **210** may include a current load of about 200 pounds and second elevator car **212** may include a current load of about 0 pounds. Additionally, first elevator car **210** may be positioned further from the first location (e.g., fourth floor **204D**) than second elevator car **220** when the call is received at step **302**. Accordingly, second elevator car **220** may be determined as an optimal elevator car from the plurality of elevator cars **210**, **220** to dispatch to fourth floor **204D**. In some embodiments, dispatch controller **130** may be configured to communicate with call device **110** to transmit a message to the prospective occupant **20** at the first location. For example, dispatch controller **130** may communicate an identification of the second elevator car **220** assigned to answer the call. In other embodiments, dispatch controller **130** may identify second elevator shaft **212** from which second elevator car

220 may arrive. The message may be transmitted via call device **110** in various suitable formats, including, for example, via a display (e.g., a written form, a graphic form, etc.), a speaker (e.g., an audible form), and more.

Dispatch controller **130** may be configured to store the sensor data **146** of each of the plurality of elevator cars **210**, **220** in memory **138**. It should be appreciated that dispatch controller **130** may continuously store sensor data **146** of elevator cars **210**, **220** in response to the repeated use of dispatch system **100** when receiving calls (step **302**) and obtaining sensor data **146** (step **304**) to determine which of the plurality of elevator cars **210**, **220** to dispatch to the call (step **308**). Accordingly, memory **138** may provide a database of load measurements for each of the plurality of elevator cars **210**, **220**. Further, dispatch controller **130** may determine a timing of when each load measurement is received by dispatch controller **130** such that the sensor data **146** stored in memory **138** may be associated with a corresponding time interval. It should be appreciated that the sensor data **146** may be accessible for review by a user of dispatch system **100** via user interface **136**.

At step **310**, dispatch controller **130** may be configured to determine a maximum load of each elevator car **210**, **220** from the one or more load measurements received from sensing devices **125** during one or more predefined periods. The predefined period may include various time intervals during which sensor data **146** is received from the plurality of elevator cars **210**, **220**. For example, the predefined period may include, but is not limited to, one or more hours of a day, one or more days of a week, one or more weeks of a month, one or more months of a year, etc. Accordingly, dispatch controller **130** may determine the maximum load measurement of each elevator car **210**, **220** for a particular predefined period. It should be appreciated that memory **138** may include corresponding load measurements (e.g., sensor data **146**) for a plurality of predefined periods.

In the present example, the predefined period may include a two-hour duration (e.g., 12:00 PM to 2:00 PM) during weekdays (e.g., Monday, Tuesday, Wednesday, Thursday, and Friday). In this instance, the maximum load measurement of each elevator car **210**, **220** may be determined from the one or more load measurements received from elevator cars **210**, **220** during the two-hour duration of each weekday. At step **314**, dispatch controller **130** may be configured to generate a modified load setting **148** for each elevator car **210**, **220** based on the maximum load received by each respective elevator car **210**, **220** during the predefined period. That is, the modified load setting **148** may be equal to the greatest load measurement received by each elevator car **210**, **220** during the predefined period.

In some embodiments, dispatch controller **130** may receive a user input (e.g., via user interface **136**), at step **312**, with a command to determine the modified load setting **148** for one or more of the plurality of elevator cars **210**, **220**. It should be appreciated that each modified load setting **148** may be associated with a particular elevator car **210**, **220** and a particular predefined period during which the maximum load measurement, from which the modified load setting **148** is derived from, was received. It should be understood that the modified load setting **148** may be applicable to the predefined period.

In the present example, the maximum load measurement received by first elevator car **210** during the predefined period may equal about 1,100 pounds, and the maximum load measurement received by second elevator car **220** during the predefined period may equal about 1,300 pounds. Accordingly, dispatch controller **130** may adjust the original

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load setting **144** of first elevator car **210** from 1,500 pounds to 1,100 pounds (e.g., the modified load setting **148**) during the two-hour duration on weekdays. Dispatch controller **130** may further adjust the load setting **144** of second elevator car **220** from 1,400 pounds to 1,300 pounds (e.g., the modified load setting **148**) during the two-hour duration on weekdays.

In other embodiments, dispatch controller **130** may be configured to automatically generate the modified load setting **148** for one or more of the plurality of elevator cars **210**, **220**. For example, dispatch controller **130** may automatically generate the modified load setting **148** based on determining the maximum load measurement is less than the load setting **144** by a predetermined threshold. The predetermined threshold may be determined by dispatch controller **130** or defined by a user of dispatch system **100**. In some examples, the predetermined threshold may range from about 5% to about 95%.

In the present example, the predetermined threshold may be set to about 20%. With the maximum load measurement of first elevator car **210** (e.g., 1,100 pounds) being less than the load setting **144** of first elevator car **210** (e.g., 1,500 pounds) by about 27%, dispatch controller **130** may automatically generate the modified load setting **148** for first elevator car **210**. Further, with the maximum load measurement of second elevator car **220** (e.g., 1,300 pounds) being less than the load setting **144** of second elevator car **220** (e.g., 1,400 pounds) by about 7%, dispatch controller **130** may forgo generating the modified load setting **148** for second elevator car **220**. It should be appreciated that dispatch controller **130** may be operable to account for small losses in load measurements attributed to various sources, including the sensing device **146**, hoist way issues, and more.

Still referring to FIG. 5, at step **314**, dispatch controller **130** may be configured to apply the modified load setting **148** in substitute of the load setting **144**. It should be understood that the modified load setting **148** may be an adjustment to the load setting **144**, and applicable in lieu of the original load setting **144**, during the predefined period. In this instance, when receiving a new call request (step **302**) during the predefined period (e.g., between 12:00 PM to 2:00 PM on weekdays), dispatch controller **130** may compare a detected load measurement of the elevator car **210**, **220** (step **304**) to the modified load setting **148** (step **306**) when determining whether the elevator car **210**, **220** includes sufficient capacity to receive the call.

Referring now to FIG. 6, an example method **400** of using dispatch system **100** to determine an occupancy at a plurality of locations, and to position inactive elevator cars at the location having a greater occupancy, is depicted. It should be understood that the steps shown and described herein, and the sequence in which they are presented, are merely illustrative such that various embodiments may include additional and/or fewer steps without departing from a scope of this disclosure. Further, it should be appreciated that dispatch system **100** may perform example method **400** in conjunction with one or more other processes, such as method **300** described above.

At step **402**, dispatch system **100** may receive a call request at a location of a plurality of locations within working environment **200**. The call may be initiated in response to a prospective occupant **20** actuating call device **110** at the location (e.g., one of floors **204A-204D**). Call device **100** may transmit the call to dispatch controller **130** via network **115**. In the present example, dispatch controller **130** may receive the call from a first location (e.g., second floor **204B**) for transportation to a second location (e.g., first

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floor **204A**). Dispatch controller **130**, in accordance with zoning logic **142**, may receive motion data from a corresponding motion controller **105** of each elevator car **210**, **220** to determine current motion parameters of the plurality of elevator cars **210**, **220**.

Dispatch controller **130** may further receive sensor data **146** from a corresponding sensing device **125** of each elevator car **210**, **220** to determine a current load of elevator cars **210**, **220**. Motion controller **105** and sensing device **125** may each transmit a signal to dispatch controller **130** (via network **115**) indicative of the motion data and the sensing data **146** of the corresponding elevator car **210**, **220**, respectively. At step **404**, dispatch controller **130** may dispatch at least one of the plurality of elevator cars **210**, **220** having a current load that does not exceed the respective load setting **144** (and/or modified load setting **148**) of the elevator car **210**, **220**, such as, for example, in accordance with the steps of method **300** described above. In the present example, first elevator car **210** may be dispatched to the first location of the call (e.g., second floor **204B**) to pick up the prospective occupant **20**.

Still referring to FIG. 6, at steps **406** to **410**, dispatch controller **130** may be configured to determine an occupancy at a plurality of locations. For example, at step **406**, dispatch controller **130** may be configured to determine a first load measurement of first elevator car **210** when arriving at the first location (e.g., a load start value). In this instance, sensing device **125** may transmit a signal to dispatch controller **130** of the first load measurement (e.g., sensor data **146**) when the motion parameters received from motion controller **105** indicate first elevator car **210** has arrived at the first location. In the present example, the first load measurement may include a load indicative of a single occupant **10** located within the cabin of first elevator car **210** when arriving at second floor **204B**.

At step **408**, dispatch controller **130** may be configured to determine a second load measurement of first elevator car **210** when departing from the first location (e.g., a load end value). In this instance, sensing device **125** may transmit a signal to dispatch controller **130** of the second load measurement (e.g., sensor data **146**) when the motion parameters received from motion controller **105** indicate first elevator car **210** has departed the first location. In the present example, the second load measurement may include a load indicative of a pair of occupants **10** located within the cabin of first elevator car **210** when departing second floor **204B**. At step **410**, dispatch controller **130** may be configured to determine a difference between the first load measurement (step **406**) and the second load measurement (step **408**) to compute a resulting occupancy at the first location. Accordingly, to determine a corresponding number of prospective occupants **10** received from (and/or transferred to) the first location, dispatch controller **130** may compare the first load measurement of first elevator car **210** when arriving at second floor **204B** to the second load measurement after departing from second floor **204B**.

In the present example, first elevator **210** may include a first load measurement of about 150 pounds to about 200 pounds upon arriving to the first location, and about 300 pounds to about 400 pounds upon departing from the first location to the destination location (e.g., first floor **204A**). Accordingly, dispatch controller **130** may be configured to determine that about one prospective occupant **20** entered first elevator car **210** from second floor **204B**. It should be appreciated that dispatch controller **130** may store a predetermined occupant load in memory **138**. In this instance, dispatch controller **130** may correlate the one or more load

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measurements to a number of occupants **10** via conversion with the predetermined occupant load. For example, the predetermined occupant load may range from about 100 pounds to about 300 pounds, such as 150 pounds. In other embodiments, the one or more load measurements may be in various other metric forms, including, for example, volts direct current (VDC). In this instance, dispatch controller **130** may correlate one volt to a predetermined load variable, such as, for example, a weight ranging from about 100 pounds to about 300 pounds. It should be appreciated that various other suitable metrics of the load measurements may be implemented by dispatch system **100** without departing from a scope of this disclosure.

Still referring to FIG. 6, at step **412**, dispatch controller **130** may determine whether the elevator car **210**, **220** is in an inactive state. For example, dispatch controller **130** may be configured to determine an operating status of first elevator car **210**, such as whether first elevator car **210** is actively completing a call and/or is assigned to answer an additional call. In the present example, upon receiving the prospective occupant **20** from second floor **204B** (e.g., the first location), first elevator car **210** may be dispatched to a destination of the prospective occupants **20** (e.g., a second location) to complete the call request. Accordingly, dispatch controller **130** may determine first elevator car **210** remains in an active state at step **412**, and return to step **404** to dispatch first elevator car **210** to the second location. Dispatch controller **130** may be configured to repeat steps **406** to **410** to determine an occupancy of the second location (e.g., first floor **204A**). Accordingly, dispatch controller **130** may measure a first load measurement of first elevator car **210** upon arriving to the second location (step **406**) and a second load measurement upon departing from the second location (step **408**). Dispatch controller **130** may determine a difference (i.e., percent load change) between the first and second load measurements (step **410**) to compute a resulting occupancy at the second location.

It should be understood that dispatch controller **130** may compute a percent load change for each of the plurality of locations when at least one of the plurality of elevator cars **210**, **220** travels to said location to answer a call (e.g., pick up a prospective occupant **20**) and/or to complete a call (e.g., drop off an occupant **10**). Dispatch controller **130** may generate local load data **150** for the first and second locations based on the occupancy computed at step **410**, respectively. The local load data **150** may include a measurement of a load transferred by first elevator car **210** to and/or from the first location (e.g., second floor **204B**) and the second location (e.g., first floor **204A**). The local load data **150** may be indicative of a number of occupants **10** located at the location after an arrival and departure of first elevator car **210** from said location.

It should be understood that the local load data **150** may include a comprehensive measurement that accounts for a cumulative load transported to, and from, the location by the plurality of elevator cars **210**, **220**. Accordingly, dispatch controller **130** may maintain a current occupancy determination for each of the plurality of locations. Dispatch controller **130** may be configured to store the local load data **150** in memory **138**, and continuously update the local load data **150** for each of the plurality of floors **204A-204D** during continued use of dispatch system **100**.

Still referring to FIG. 6, at step **412**, dispatch controller **130** may determine whether the elevator car **210**, **220** is in an inactive state. Dispatch controller **130** may determine that first elevator car **210** is in an inactive state when no further calls are assigned to first elevator car **210**, and/or first

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elevator car **210** does not include additional destinations from existing calls. In response to determining first elevator car **210** is in an inactive state at step **412**, dispatch controller **130** may be configured to determine at least one location of the plurality of locations that has a maximum occupancy at step **414**. That is, dispatch controller **130** may be configured to compare the local load data **150** of the plurality of locations relative to one another to assess a current occupancy at each location.

Dispatch controller **130** may determine fourth floor **204D** includes an occupancy that is greater than the occupancy of the remaining plurality of locations. In the present example, as seen in FIG. 2, first floor **204A** may include two occupants **20** (e.g., recently transported thereto by first elevator car **210**), second floor **204B** may include one remaining occupant **20**, third floor **204C** may include two occupants **20**, and fourth floor **204D** may include three occupants **20**. Accordingly, dispatch controller **130** may determine that fourth floor **204D** includes a current occupancy that is greater than the occupancy of the remaining floors **204A-204C**.

At step **416**, dispatch controller **130** may be configured to move first elevator car **210** to fourth floor **204D**. First elevator car **210** may be positioned at fourth floor **204D** while first elevator car **210** remains in an inactive state. Stated differently, first elevator car **210** may be parked at fourth floor **204D** until a call request from one of the plurality of floors **204A-204D** (e.g., via call device **110**) is assigned to first elevator car **210** by dispatch controller **130**. It should be appreciated that, with first elevator car **210** maintained at fourth floor **204D**, and with fourth floor **204D** including a greater occupancy than the remaining plurality of floors **204A-204C**, a minimum travel distance for answering a future call request with first elevator car **210** may be minimized.

It should be appreciated that dispatch controller **130** may be configured to periodically reassess the current occupancy (e.g., local load data **150**) of each of the plurality of floors **204A-204D**. Accordingly, dispatch controller **130** may move one or more inactive elevator cars **210**, **220** to a modified location based on updated local load data **150**. For example, in response to determining the first location (identified at step **414**) no longer includes a greater occupancy relative to the plurality of other locations, dispatch controller **130** may be configured to reposition the inactive elevator car(s) **210**, **220** to a second location having the greatest occupancy.

In some embodiments, method **300** may include further steps for positioning one or more inactive elevators at additional locations when a number of inactive elevator cars **210**, **220** at the first location (e.g., fourth floor **204D**) exceeds a predetermined threshold. In other embodiments, a user of dispatch system **100** may identify a number of locations at which the plurality of elevator cars **210**, **220** may be parked at when in an inactive state. For example, dispatch controller **130** may receive a user input (e.g., via user interface **136**) indicating three locations for parking inactive elevator cars **210**, **220**. In this instance, dispatch controller **130** may determine which three locations of the plurality of locations have the greatest occupancy relative to the remaining plurality of locations, and direct any inactive elevator cars **210**, **220** to at least one of the three locations. In some embodiments, dispatch controller **130** may be operable to generate a report (e.g., via user interface **136**) including information relating to one or more of the load setting **144**, the sensor data **146**, the modified load setting **148**, the local load data **150**, and more.

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It should be appreciated that the one or more processes of dispatch system **100** shown and described herein, such as example methods **300**, **400**, may be implemented in various other working environments. In one example, dispatch system **100** may be configured to apply one or more of example methods **300**, **400** in a transit system, such as a bus service, a train service, a subway service, a metro service, a ride-sharing service, etc. With respect to example method **300**, dispatch system **100** may render a transportation unit (e.g., a bus, a train, a subway, a metro, a vehicle, etc.) inoperable for receiving additional calls and/or occupants when exceeding its maximum load capacity. In this instance, the transportation unit may bypass the location (e.g., the stop) and/or inhibit receipt of additional load onto the transportation unit (e.g., by not opening doors). In some embodiments, dispatch system **100** may be configured to communicate with one or more remote stations to transmit information indicative of a current load.

For example, dispatch system **100** may transmit alerts to remote station(s) requesting assistance from additional transportation units (e.g., a bus, a train, a subway, a metro, a vehicle, etc.) at one or more locations when the current load of one or more current transportation units exceed a maximum load capacity. It should be appreciated that dispatch system **100** may promote traffic flow by determining a minimum number of transportation units required at one or more locations, or at one or more predefined intervals, to accommodate an expected load based on local load data of various locations. With respect to example method **400**, dispatch system **100** may determine an occupancy at a plurality of locations (e.g., bus stops, train stops, subway stops, metro stops, etc.) to position inactive transportation unit (e.g., a bus, a train, a subway, a metro, a vehicle, etc.) at the location having a greater occupant count.

All technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this disclosure belongs unless clearly indicated otherwise. As used herein, the singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

The above description is illustrative and is not intended to be restrictive. One of ordinary skill in the art may make numerous modifications and/or changes without departing from the general scope of the disclosure. For example, and as has been described, the above-described embodiments (and/or aspects thereof) may be used in combination with each other. Additionally, portions of the above-described embodiments may be removed without departing from the scope of the disclosure. In addition, modifications may be made to adapt a particular situation or material to the teachings of the various embodiments without departing from their scope. Many other embodiments will also be apparent to those of skill in the art upon reviewing the above description.

What is claimed is:

1. A method of operating an elevator car, comprising:
 - receiving one or more load measurements of the elevator car during a plurality of predefined periods when the elevator car is operational;
 - determining a maximum load of the elevator car that corresponds to each of the plurality of predefined periods from the one or more load measurements received during each of the plurality of predefined periods;
 - generating a modified load setting for the elevator car for each of the plurality of predefined periods based on the

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maximum load determined for the elevator car during each of the plurality of predefined periods; and replacing a load setting of the elevator car during an occurrence of at least one of the plurality of predefined periods with the modified load setting generated for the at least one predefined period, wherein the modified load setting replaces the load setting for a duration of the at least one predefined period such that the elevator car is operated in accordance with the modified load setting generated for the duration of the at least one predefined period until completion of the at least one predefined period.

2. The method of claim 1, wherein prior to generating the modified load setting for the elevator car for the at least one predefined period, the method further comprises:

- comparing the maximum load of the elevator car determined for the at least one predefined period to the load setting;

- determining a difference between the maximum load for the at least one predefined period to the load setting originally associated with the elevator car; and generating the modified load setting when the difference between the maximum load and the load setting exceeds a predetermined threshold.

3. The method of claim 2, further comprising:

- foregoing generating the modified load setting for the elevator car for the at least one predefined period when the difference between the maximum load and the load setting does not exceed the predetermined threshold, such that the elevator car is operated in accordance with the load setting during the occurrence of the at least one predefined period.

4. The method of claim 1, wherein determining the maximum load of the elevator car that corresponds to each of the plurality of predefined periods comprises:

- determining a plurality of maximum loads for the plurality of predefined periods with at least one maximum load corresponding to each predefined period of the plurality of predefined periods.

5. The method of claim 1, further comprising:

- receiving a new load measurement associated with the elevator car during the at least one predefined period; comparing the new load measurement to the modified load setting corresponding to the at least one predefined period;

- rendering the elevator car inoperable to receive a call in response to the new load measurement received during the at least one predefined period exceeding the modified load setting that corresponds to the at least one predefined period; or

- rendering the elevator car operable to receive the call in response to the new load measurement received during the at least one predefined period not exceeding the modified load setting that corresponds to the at least one predefined period.

6. The method of claim 1, further comprising:

- replacing the modified load setting with the load setting upon completion of the duration of the at least one predefined period, such that the elevator car is operated in accordance with the load setting until commencement of another one of the plurality of predefined periods.

7. The method of claim 6, further comprising:

- receiving a new load measurement associated with the elevator car after completion of the at least one predefined period;

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comparing the new load measurement to the load setting that replaced the modified load setting; and rendering the elevator car inoperable to receive a call in response to the new load measurement received after the at least one predefined period exceeding the load setting; or
rendering the elevator car operable to receive the call in response to the new load measurement received after the at least one predefined period not exceeding the load setting.

8. The method of claim 1, wherein the one or more load measurements are received during each of the plurality of predefined periods, such that the maximum load of the elevator car for each of the plurality of predefined periods is the load measurement that equals a greatest load associated with the elevator car during each of the plurality of predefined periods.

9. The method of claim 1, wherein the one or more load measurements of the elevator car defines a total load of the elevator car including a number of occupants and a number of ancillary objects located within the elevator car.

10. The method of claim 1, wherein the load setting defines a first maximum load capacity of the elevator car, and the modified load setting for each of the plurality of predefined periods define a second maximum load capacity of the elevator car that is less than the first maximum load capacity, such that the modified load setting across each of the plurality of predefined periods are all less than the load setting.

11. The method of claim 1, wherein the plurality of predefined periods includes predefined time intervals throughout the duration, including:
hourly intervals throughout a day;
daily intervals throughout a week;
weekly intervals throughout a month; or
monthly intervals throughout a year.

12. A method of operating a plurality of elevator cars, comprising:

measuring at least one load of each of the plurality of elevator cars during multiple predefined periods;
determining a maximum load of each of the plurality of elevator cars from the at least one load measured during each of the multiple predefined periods;
generating a modified load setting associated with each of the multiple predefined periods for each of the plurality of elevator cars, such that the modified load setting associated with each of the multiple predefined periods is based on the maximum load determined for each respective elevator car of the plurality of elevator cars during an occurrence of each of the multiple predefined periods; and

activating the modified load setting for one or more elevator cars of the plurality of elevator cars, during the occurrence of at least one of the multiple predefined periods, in place of a load setting originally associated with the one or more elevator cars, such that the one or more elevator cars are operated in accordance with the modified load setting generated for the at least one of the multiple predefined periods until completion of the at least one of the multiple predefined periods;
wherein the modified load setting define an adjusted capacity of each of the plurality of elevator cars relative to the load setting originally associated with each of the plurality of elevator cars.

13. The method of claim 12, further comprising:
receiving a call for the one or more elevator cars during the at least one of the multiple predefined periods;

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measuring a new load for the one or more elevator cars in response to receiving the call and comparing the new load to the modified load setting of the one or more elevator cars that corresponds to the at least one of the multiple predefined periods;

determining the new load of the one or more elevator cars does not exceed the modified load setting of the one or more elevator cars and rendering the one or more elevator cars operable to receive the call, such that at least one of the one or more elevator cars is dispatched to the call; or

determining the new load of the one or more elevator cars does exceed the modified load setting of the one or more elevator cars and rendering the one or more elevator cars inoperable to receive the call, such that the one or more elevator cars are prevented from being dispatched to the call.

14. The method of claim 12, wherein the load setting is substituted by the modified load setting during the occurrence of at least one of the multiple predefined periods, the method further comprising:

replacing the modified load setting with the load setting upon completion of the at least one of the multiple predefined periods.

15. The method of claim 12, wherein prior to generating the modified load setting of a first elevator car of the plurality of elevator cars for at least one of the multiple predefined periods, the method further comprises:

comparing the maximum load of the first elevator car for the at least one predefined period to the load setting originally associated with the first elevator car;
determining a difference between the maximum load and the load setting; and
generating the modified load setting when the difference between the maximum load and the load setting exceeds a predetermined threshold.

16. The method of claim 15, further comprising:
foregoing generating the modified load setting for the first elevator car during the at least one predefined period when the difference between the maximum load and the load setting does not exceed the predetermined threshold, such that the first elevator car is operated in accordance with the load setting during the occurrence of the at least one predefined period.

17. The method of claim 12, wherein the multiple predefined periods includes predefined time intervals throughout a duration, including:

hourly intervals throughout a day, wherein the hourly intervals are at least one hour durations such that the load setting is replaced with the modified load setting during the at least one hour duration of the hourly intervals.

18. A system for operating an elevator car, comprising:
a processor; and
a memory storing instructions that, when executed by the processor, causes the processor to perform operations including:

measuring one or more loads of the elevator car during a plurality of predefined periods;
determining a maximum load of the elevator car that corresponds to each of the plurality of predefined periods from the one or more loads measured during each of the plurality of predefined periods;
generating a modified load setting for the elevator car for each of the plurality of predefined periods based

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on the maximum load determined for the elevator car during each of the plurality of predefined periods; and

replacing a load setting of the elevator car during an occurrence of at least one of the plurality of predefined periods with the modified load setting generated for the at least one predefined period, wherein the modified load setting replaces the load setting for a duration of the at least one predefined period such that the elevator car is operated in accordance with the modified load setting generated for the duration of the at least one predefined period until completion of the at least one predefined period.

19. The system of claim **18**, wherein prior to generating the modified load setting for the elevator car for the at least one predefined period, the processor further performing operations including:

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comparing the maximum load of the elevator car that corresponds to each of the plurality of predefined periods to the load setting;

determining a difference between the maximum load for each of the plurality of predefined periods to the load setting;

generating the modified load setting for one or more of the plurality of predefined periods when the difference between the maximum load and the load setting exceeds a predetermined threshold; and

foregoing generating the modified load setting for the one or more plurality of predefined periods when the difference between the maximum load and the load setting does not exceed the predetermined threshold, such that the elevator car is operated in accordance with the load setting during the occurrence of the one or more plurality of predefined periods.

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